# MARTIJN IJTSMA

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## EDUCATION

**Doctor of Philosophy in Aerospace Engineering** August 19th, 2016 - August 3rd, 2019
*Georgia Institute of Technology, United States*

- Dissertation title: "Computational Simulation of Adaptation of Work Strategies in Human-Robot Teams"
- Advisor: Dr. Amy Pritchett
- Committee: Dr. Karen Feigh, Dr. Matthew Johnson, Dr. John-Paul Clarke, Dr. Glenn Lightsey,
- Coursework focused on systems design, optimization, flight mechanics, and control
- Certificate program (Tech to Teaching) for preparing future faculty and develop teaching skills

**Master of Science in Aerospace Engineering** September 2013 - June 2016
*Delft University of Technology, Netherlands*

- Thesis title: "Adaptive Automation Based on Air Traffic Controller's Decision-Making"
- Advisor: Dr. ir. Clark Borst
- Committee: Dr. ir. René van Paassen, Dr. ir. Joost de Winter, Dr. ir. Max Mulder, Ir. Gustavo Mercado-Velasco
- Coursework focused on control and simulation of aerospace vehicles

**Bachelor of Science in Aerospace Engineering** September 2010 - July 2013
*Delft University of Technology, Netherlands*

## EXPERIENCE

**Associate Professor in Human Systems Integration** June 2026 - Present
*Department of Integrated Systems Engineering, The Ohio State University*
*Department of Mechanical and Aerospace Engineering, The Ohio State University*

- My research and teaching aims to improve operations involving humans and technology working together in naturalistic, high-complexity environments. This includes the study of human-AI/robot systems and distributed work in high-stakes applications. I specifically innovate new approaches for the design and evaluation of joint human-machine systems by integrating techniques from cognitive work/task analysis, multi-agent modeling & simulation, control theory, systems theory, and field research. Current domains of practice include air traffic management, disaster response, spaceflight operations, and smart mobility, among others.
- I lead a team of graduate and undergraduate research assistants as part of the Cognitive Systems Engineering Lab (CSEL).
- **External Funding & Awards**
  - NASA: Cognitive Systems Engineering (CSE) Methods for Support Adaptive, Integrated Anomaly Response Phase-2. Status: *In Progress.* **$250k to OSU**
  - Computational Human Interoperation with Machine Partners (CHIMP). Status: *In Progress.* **$150k to OSU**
  - NASA: Cognitive Systems Engineering (CSE) Methods for Support Adaptive, Integrated Anomaly Response Phase-1. Status: *Completed.* **$45k to OSU**
  - National Science Foundation: CAREER: Making Robots More Cooperative Agents: Controlling Costs of Coordination Through Graph-Based Models of Joint Activity. Status: *In Progress.* **$554k to OSU**
  - Federal Aviation Administration: Pilot Interactions with Advanced Flight Deck Technologies. Status: *In Progress.* **$1m to OSU**

- NASA: Contingency Planning Toolkit for Advanced Air Mobility Phase 2. Status: *Completed*. $140k total award, **$163k to OSU**
- 99P Labs: Designing for Human-AI Teaming in Smart Mobility. Status: *Completed*. **$162k to OSU**
- Federal Aviation Administration: Reliance on Automated or Complex Flight Deck Systems in Commercial Aircraft: Evaluating Compliance to 14 CFR 25.1302(c) and 25.1329(i). Status: *Completed*. **$90k to OSU**
- NASA: Contingency Planning Toolkit for Advanced Air Mobility Phase 1. Status: *Completed*. $140k total award, **$42k to OSU**
- Ohio Federal Research Network: Interoperability, Resiliency, and Contingency Management for Ohio UAS Operations. Status: *Completed*. $1.4M total award, **$150k to OSU**
- Phase I STTR AFWERX Agility Prime: Rapid Mission Planner for UAV Operations. Status: *Completed*. $150k total award, **$45k to OSU**

- **M.S. and Ph.D. Advising**
  - Emily Barrett – M.S. Integrated Systems Engineering. Graduated August 2021
  - Stephanie Duros – M.S. Integrated Systems Engineering. Graduated August 2022
  - Kathleen Albert – M.S. Integrated Systems Engineering. Graduated December 2022
  - Jacob Keller – M.S. Integrated Systems Engineering. Graduated May 2023
  - Renske Nijveldt – M.S. Integrated Systems Engineering. Graduated May 2023
  - Kenneth Cassidy – M.S. Integrated Systems Engineering, Graduated May 2023
  - Joan Smith – M.S. Integrated Systems Engineering, Graudated May 2024
  - Abhinay Paladugu – Ph.D. Integrated Systems Engineering, Graduated August 2024
  - Abigail Post – Ph.D. Integrated Systems Engineering
  - Jennifer Winner – Ph.D. Integrated Systems Engineering
  - Connor Kannally – Ph.D Integrated Systems Engineering
  - Sal Hargis – Ph.D. Integrated Systems Engineering

- **Teaching**
  - ISE 3700 – Introduction to Cognitive Systems Engineering (SP20, SP21, SP22, SP23, SP24)
  - ISE 7720 – Cognitive Systems Engineering: Models and Methods (AU20, AU21, AU22, AU24)
  - ISE 5740 – Human-Centered Automation (AU21, AU22, AU23)
  - ISE 7700 – Advanced Topics in Cognitive Systems Engineering (SP25, SP26)
  - ISE 5194 – Workshop on Interdisciplinary Discourse and Emerging Technology (SP26)

**Assistant Professor in Human Systems Integration** September 2019 - June 2026
*Department of Integrated Systems Engineering, The Ohio State University*
*Department of Mechanical and Aerospace Engineering, The Ohio State University*

**Graduate Research Assistant** August 2016 - August 2019
*Cognitive Engineering Center, Georgia Institute of Technology*

- **Projects**
  - NASA Human Research Program: Objective Function Allocation Method for Human-Automation/Robotic Interaction using Work Models that Compute
  - NASA Space Technology Research Grant: Technologies for Mixed-Initiative Plan Management for Human Space Flight

**Research Intern** October 2014 - March 2015
*Cognitive Engineering Center, Georgia Institute of Technology*

- **Projects**
  - NASA Aviation Safety Program: Scenario Based Methods for Verification of Authority and Autonomy

**Teaching Assistant** November 2012 - July 2014
*Delft University of Technology, Netherlands*

· Taught in recital-style classrooms, mentored student groups in project-based learning, and graded exams and project assignments.

  · Dynamics, AE1130-II, Dr. Sergio Turteltaub
  · Aerospace System Design, AE2111-I, Dr. Nando Timmer
  · Test, Analysis & Simulation, AE2223-I, Dr. Mirjam Snellen

**Mathematics and Chemistry Teacher** December 2011 - September 2015
*Stichting Studiebegeleiding Leiden, Netherlands*

· Mentored high school students in preparation for their final exams, collaboration with the University of Leiden.

## PUBLICATIONS

### Journal Articles

10. **IJtsma, M.** & Hargis, S. (2026). Envisioning the Work of Reconfiguring: A Modeling Perspective for Designing Adaptive Human-Agent Systems. *AI Magazine, 47*(3). https://doi.org/10.1002/aaai.70085

9. Post, A., Nijveldt, R., Woods, D., & **IJtsma, M.** (2026). Determining What's Next: Visual Analytics for Evaluating Human-Automation Coordination. *Cognition, Technology & Work, 68*(1). https://doi.org/10.1177/10711813241276459

8. Paladugu, A., Fernandes, A., & **IJtsma, M.** (2025). Computational Simulation of Distributed Work as a Discovery Tool for Envisioning Future Operations. *Journal of Cognitive Engineering and Decision Making, 19*(4). https://doi.org/10.1177/15553434251327698

7. Kannally, C., Paladugu, A., Nijveldt, R., McSherry, L. & **IJtsma, M.** (2025). An Exploratory Study of Contextual Control Modes in Teamwork. *Human Factors, 67*(5). https://doi.org/10.1177/00187208241292669

6. Barrett, E., Paladugu, A., & **IJtsma, M.** (2024). Multi-Agent Simulation to Envision Communication Strategies in Future Air Mobility Operations. *AIAA Journal of Aerospace Information Systems, 21*(8), 605-615. https://doi.org/10.2514/1.I011239

5. Keller, J.R., **IJtsma, M.**, Newton, E.K. (2023). Examining autonomous flight safety systems from a cognitive systems engineering perspective: Challenges, themes, and outlying risks. *Journal of Space Safety Engineering, 10*(1), 76-81. https://doi.org/10.1016/j.jsse.2022.11.005

4. **IJtsma, M.**, Borst, C., Mulder, M., & Van Paassen, M.M. (2022). Evaluation of a Decision-Based Invocation Strategy for Adaptive Support for Air Traffic Control. *IEEE Transactions on Human-Machine Systems, 52*(6), 1135-1146. https://doi.org/10.1109/THMS.2022.3208817

3. Ma, L. M., **IJtsma, M.**, Feigh, K. M., and Pritchett, A. R. (2022). Metrics for Human-Robot Team Design: A Teamwork Perspective on Evaluation of Human-Robot Teams. *ACM Transactions on Human-Robot Interaction, 11*(3). 1-36. https://doi.org/10.1145/3522581

2. **IJtsma, M.**, Ma, L.M., Pritchett, A.R., & Feigh, K.M. (2019). Computational Methodology for the Allocation of Work and Interaction in Human-Robot Teams. *Journal of Cognitive Engineering and Decision Making, 13*(4), 221-241. https://doi.org/10.1177/1555343419869484

1. Pritchett, A. R., Bhattacharyya, R. P., & **IJtsma, M.** (2016). Computational Assessment of Authority and Responsibility in Air Traffic Concepts of Operation. *Journal of Air Transportation, 24*(3), 93-101. https://doi.org/10.2514/1.D0024

### Journal Manuscripts in Review / Preprints

3. **IJtsma, M.**, & Hargis, S. (In Review). A Graph-Based Conceptual Modeling Framework to Design for Human–Robot Coordination. *Submitted to IEEE Transactions on Human-Machine Systems.* https://doi.org/10.48550/arXiv.2512.15282

2. Kannally, C. & **IJtsma, M.** (In Revision). Locally Adaptive, Globally Maladaptive: Exploring Sensemaking Dynamics in Distributed Work Using Work Models that Compute. *Submitted to Journal of Cognitive Engineering and Decision Making.*

1. Paladugu, A., Fernandes, A., & **IJtsma, M.** (In Review). Synchronizing Multi-Threaded Activity under Pressure: A Simulation Study of Contingency Response in Envisioned Air Traffic Management Operations. *Submitted to Cognition, Technology & Work.* https://doi.org/10.21203/rs.3.rs-9839654/v1

**Journal Manuscripts in Preparation**

**Book Chapters**

1. **IJtsma, M.**, Ma, L.M., Feigh, K.M., & Pritchett, A.R. (2019). Analysis of Work Dynamics for Objective Function Allocation in Manned Spaceflight Operations. In M.A. Vidulich & P. Tsang (Eds.), *Improving Aviation Performance through Applying Engineering Psychology, Advances in Aviation Psychology.*

**Technical Reports**

1. Post, A.M., Nijveldt, R., Woods, D.D., Pruchnicki, S., & **IJtsma, M.** (2025). *Report on the Reliance on Automated or Complex Flight Deck Systems in Commercial Aircraft: Evaluating Compliance to 14 CFR 25.1302(c ) and 25.1329(i)* (Report No. 2025-001). The Ohio State University Cognitive Systems Engineering Lab.

**Conference Proceedings**

*Fully Reviewed*

5. Hargis, S., Kannally, C., & **IJtsma, M.** (2025). A Driver-in-the-Loop Study of Human-AI Joint Replanning. Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Chicago.

4. **IJtsma, M.**, Lassiter, W., Feigh, K.M., Savelsbergh, M., & Pritchett, A.R. (2019). *An Integrated System for Mixed-Initiative Planning of Manned Spaceflight Operations.* Paper presented at the 2019 IEEE Aerospace Conference, Big Sky, MT.

3. Baltrusaitis, M., Feigh, K.M., **IJtsma, M.**, Lassiter, W., Pritchett, A.R., & Savelsbergh, M. (2018). *Technologies for Mixed-Initiative Plan Management for Human Space Flight.* Paper presented at the International Conference on Automated Planning and Scheduling, Delft, Netherlands.

2. Ma. L.M., **IJtsma, M.**, Feigh, K.M., Paladugu, A., & Pritchett, A.R. (2018). *Modelling and Evaluating Failures in Human-Robot Teaming Using Simulation.* Paper presented at the 2018 IEEE Aerospace Conference, Big Sky, MT.

1. **IJtsma, M.**, Bhattacharyya, R.P., Pritchett, A.R., & Hoekstra, J. (2015). *Computational Assessment of Different Air-Ground Function Allocations.* Paper presented at the 11th USA/Europe Air Traffic Management Research and Development Seminar, Lisbon, Portugal.

*Abstract Reviewed*

23. Paladugu, A., Post, A., & **IJtsma, M.** (2025). Two Tools for Analyzing Coordination Problems To Design for Distributed Work: A Case Study in Urban Air Mobility. Paper presented at the 23rd International Symposium on Aviation Psychology, Oregon.

22. Fernandes, A., Zabara, K., Epps, K., **IJtsma, M.**, Paladugu, A., & Calhoun, S. (2025). *Use of Model Based System Engineering to Drive UAM Contingency Management Procedure Design.* Paper presented at AIAA Scitech 2025 Forum, Orlando, FL.

21. Post, A., Nijveldt, R., Woods D.D. & **IJtsma, M.** (2024). *Determining What's Next: Visual Analytics for Evaluating Human-Automation Coordination.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Phoenix.

20. Paladugu, A., Fernandes, A. & **IJtsma, M.** (2024). *The Use of Computational Modeling and Simulation to Design and Evaluate a Distributed Work System.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Phoenix.

19. Hargis, S. & **IJtsma, M.** (2024). *Functional Models of Work to Examine Human-Robot Coordination in Disaster Response.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Phoenix.

18. Kannally, C., Smith, J. & **IJtsma, M.** (2023). *Human-AI Teaming in the Automotive and Mobility Industry: Guiding Design to Support Joint Activity.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Washington DC.

17. Fernandes, A., Wilson, S., **IJtsma, M.**, Paladugu, A., Davis, T. & Lichty, J. (2023). *Contingency Planning Toolkit for Emerging Air Mobility Ecosystems.* In AIAA AVIATION 2023 Forum (p. 3552). San Diego

16. Paladugu, A., Fernandes, A., Wilson, S., Davis, T., Lichty, J., **IJtsma, M.** (2023). *Evaluating Envisioned Air Mobility Architectures Using Computational Simulations of Work.* Paper presented at the International Symposium on Aviation Psychology. Rochester, NY.

15. **IJtsma, M.** (2022). *Situated Work in Teams: Modeling Coordination through Extending Strategies Analysis and Contextual Control.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Atlanta, GA.

14. Paladugu, A., Nijveldt, R., Cassidy, K., & **IJtsma, M.** (2022). *Strategy Selection in Teams: Exploring How Teams Coordinate Responses to Time Pressure.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Atlanta, GA.

13. Nijveldt, R., & **IJtsma, M.** (2022). *Cognitive Task Analysis of Contingency Management in Future Unmanned Aircraft Systems Traffic Management.* In AIAA AVIATION 2022 Forum (p. 3620). Chicago

12. Duros, S., Lo, J., Cassidy, K., & **IJtsma, M.** (2022). *Development of a Dynamic Model of Adaptation in Distributed Work Systems.* Paper presented at AIAA Scitech 2022 Forum.

11. Keller, J., **IJtsma, M.**, & Newton, E. K. (2021). A Critical Examination of Autonomous Flight Safety Systems from a Cognitive Systems Engineering Perspective: Challenges, Themes, and Outlying Risks. 72nd International Astronautical Congress (IAC) Proceedings, Dubai, United Arab Emirates

10. Keller, J., & **IJtsma, M.** (2021). *Requirements for Computational Approaches to Analyzing Resilience in Human-Machine Teams.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.

9. Albert, K., & **IJtsma, M.** (2021). *Modeling the Effects of Machine Rigidities on Joint Work Strategies.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.

8. Barrett, E., **IJtsma, M.** (2021). *Modeling Contingency Management in Unmanned Aircraft Systems Traffic Management.* Paper presented at the International Symposium on Aviation Psychology, Corvallis, OR.

7. Ma, L., Ye, S., **IJtsma, M.**, Feigh, K.M., & Pritchett, A.R. (2020). *An Experimental Refinement of Computational Models of Human-Robot Teams.* Paper presented at AIAA Scitech 2020 Forum.

6. **IJtsma, M.**, Ye, S., Feigh, K.M. & Pritchett, A.R. (2019). *Simulating Human-Robot Teamwork Dynamics for Evaluation of Work Strategies in Human-Robot Teams.* Paper presented at the International Symposium on Aviation Psychology, Dayton, OH.

5. **IJtsma, M.**, Ma, L.M., Feigh, K.M., & Pritchett, A.R. (2018). *Demonstration of the "Work Models that Compute" Simulation Framework for Objective Function Allocation.* Paper presented at the Human Factors and Ergonomics Society Annual Meeting, Philadelphia, PA.

4. **IJtsma, M.**, Pritchett, A.R., Ma, L.M., & Feigh, K.M. (2017). *Modeling Human-Robot Interaction to Inform Function Allocation in Manned Spaceflight Operations.* Paper presented at Robotics: Science and Systems, Boston, MA.

3. **IJtsma, M.**, Borst, C., Mercado-Velasco, G.A., Mulder, M., & Van Paassen, M.M. (2017) *Adaptive Automation Based on Air Traffic Controller's Decision-Making.* Paper presented at the International Symposium on Aviation Psychology, Dayton, OH.

2. **IJtsma, M.**, Ma, L.M., Pritchett, A.R., & Feigh, K.M. (2017). *Work Dynamics of Taskwork and Teamwork in Function Allocation for Manned Spaceflight Operations.* Paper presented at the International Symposium on Aviation Psychology, Dayton, OH.

1. **IJtsma, M.**, Bhattacharyya, R.P., & Pritchett, A.R. (2015). *Computational Simulation of Authority-Responsibility Mismatches in Air-Ground Function Allocation.* Paper presented at the International Symposium on Aviation Psychology, Dayton, OH.

**Theses**

2. **IJtsma, M.** (2019). *Computational Simulation of Adaptation of Work Strategies in Human-Robot Teams* (doctoral thesis). Georgia Institute of Technology, United States

1. **IJtsma, M.** (2016). *Adaptive Automation Based on Air Traffic Controller's Decision-Making* (master's thesis). Delft University of Technology, Netherlands

## INVITED TALKS

- "Modeling and Simulation of Joint Activity to Support Human and Machine Coordination," Rutgers University Department of Industrial Systems Engineering, October 2023
- "Making Machines More Cooperative Agents: Formative Models to Guide Human-Machine System Design," AIAA Intelligent Systems workshop, Boulder, CO, July 2023
- "Making Robots More Cooperative Agents: Evaluating Human-Robot System Designs from a Teamwork Perspective," Aerospace Human Factors New and Emerging Technologies Seminar Series, Federal Aviation Administration, May 2023
- "The Choreography of Work: Computational Modeling to Analyze and Support Coordination in Human-Machine Systems," Southern Ohio Chapter of the Human Factors and Ergonomics Society (HFES), November 2021
- "The Choreography of Work: Computational Modeling to Examine and Support Coordination in Human-Machine Systems," Department of Mechanical and Aerospace Engineering, The Ohio State University, November 2021
- "Computational Models of Work to Study Human-Robot Teaming", Aptima Human-Centered Engineering, February 2020
- "Designing Distributed Cognitive Work Systems", IISE Student Chapter General Body Meeting, The Ohio State University, February 2020
- Aerospace SCOPE Meeting, Battelle Center for Science, Engineering and Public Policy, The Ohio State University, February 2020
- "Predicting Dynamics of Human-Automation Interaction Using Computational Models of Work," Center for Automotive Research (CAR), The Ohio State University, January 2020
- "Computational Models of Work to Study Multi-Agent Sociotechnical System," Department of Integrated Systems Engineering, The Ohio State University, November 2018.

## PROFESSIONAL MEMBERSHIPS

**Human Factors and Ergonomics Society (HFES)** June 2018 - Present
**American Institute of Aeronautics and Astronautics (AIAA)** December 2020 - Present

## HONORS, AWARDS, AND CERTIFICATES

**NSF CAREER Award** February 2023
**The Stanley Nelson Roscoe Best Student Paper Award** May 2017
**Graduated Cum Laude, MSc degree** June 2016

**Graduated Cum Laude, BSc degree** July 2013
**Associate Certification of the Center for the Integration of Research, Teaching and Learning** May 2019